

PageRank Feasibility Audit of Wikipedia RfA Support Votes

Formal Project Report

ADAN7905 - AI / ML Software Tools and Platforms

Dataset: SNAP Wikipedia Requests for Adminship (RfA)

Generated from final pipeline outputs

Project Snapshot	Value
Clean vote records	196,511
Directed support edges	139,272
Support-only graph nodes	10,004
PageRank-uphill support edges	103,065 (74.00%)
Best predictive model	Random forest (F1 = 0.9496, ROC-AUC = 0.9783)
Signed PageRank user universe	11,377 users across support, oppose, and neutral layers

Executive Summary

This report documents a reproducible data science mini-project that studies PageRank behavior in the Stanford Network Analysis Project Wikipedia Requests for Adminship dataset. The project begins with a support-only directed graph in which nodes are Wikipedia users and directed edges represent support votes from voters to candidates. It computes PageRank, labels support edges as PageRank-uphill or non-uphill, engineers local graph features, and trains machine-learning models to predict whether a support edge points toward a higher-PageRank target.

The cleaned dataset contains 196,511 usable vote records, including 143,822 support votes, 40,637 oppose votes, and 12,052 neutral votes. The support-only graph contains 10,004 nodes and 139,272 directed support edges. The central empirical pattern is that 103,065 support edges (74.00%) are PageRank-uphill, meaning they point from a lower-PageRank source to a higher-PageRank target. This indicates that support in the RfA network typically flows toward structurally central candidates.

Machine-learning results show that this uphill behavior is predictable from local graph features. The random forest classifier performs best with accuracy 0.9251, F1 0.9496, and ROC-AUC 0.9783, outperforming both the majority-class baseline and logistic regression. The project also includes a lifted signed PageRank component that incorporates support and oppose votes through positive and negative reputation states while treating neutral votes as a separate attention layer. This signed extension expands the analysis from positive support centrality to a multidimensional interpretation of support, opposition, neutral attention, total signed attention, and net signed reputation.

1. Research Question and Analytical Objective

Primary Research Question

Do Wikipedia RfA support-vote edges tend to point toward users with higher PageRank, and can this PageRank-uphill behavior be predicted from local graph features?

Each directed support edge is interpreted as:

```
source voter -> target candidate
```

An edge is labeled PageRank-uphill if:

```
PageRank(target) > PageRank(source)
```

This converts the graph-theoretic PageRank question into an edge-level supervised classification problem. The outcome variable is whether a support vote points toward a structurally more central target.

Signed-Network Question

How does the centrality interpretation change when support, oppose, and neutral votes are incorporated into the analysis rather than using support votes alone?

The signed-network component addresses this question by computing a lifted signed PageRank model. Instead of treating opposition as impossible negative probability, the method represents each user with a positive and negative reputation state. Support edges preserve sign, oppose edges flip sign, and neutral votes are modeled as a separate neutral-attention PageRank layer.

2. Dataset and Data Quality

The primary dataset is the SNAP Wikipedia Requests for Adminship dataset. It is a directed, signed voting network in which nodes represent Wikipedia users and edges represent votes in the Request for Adminship process. The raw fields include source user, target user, vote sign, result, year, date/time, and vote comment text.

Raw Field	Meaning
SRC	Source user / voter
TGT	Target user / candidate
VOT	Vote sign: support, neutral, or oppose
RES	Result of the RfA nomination
YEA	Year
DAT	Date/time
TXT	Vote comment text

Dataset Strengths

- It is a real-world public research dataset from a legitimate network science source.
- It is directed, which is required for PageRank and edge-flow interpretation.
- It is signed, allowing support, neutral, and oppose votes to be distinguished.
- It is large enough to support graph analytics, supervised learning, signed-network analysis, and visualization.
- The voting process is socially interpretable because edges represent actual RfA voting behavior.

Dataset Weaknesses and Handling Choices

- The dataset is historical and covers RfA activity from 2003 through May 2013, so the results should not be assumed to describe current Wikipedia behavior.
- Vote text is retained in the cleaned table but not used in the main predictive model.
- PageRank results depend on graph construction choices, including duplicate aggregation, damping factor, weighting, and whether the graph is treated as unsigned or signed.
- The model describes observed structural patterns, not causal social mechanisms.
- Wikipedia usernames may contain spaces, punctuation, non-English characters, or URL-encoded strings. These values were retained as valid user identifiers unless the source or target field was blank or missing.

3. Reproducible Workflow and Repository Design

The project is organized as a modular pipeline so that each stage can be rerun locally or inside Docker. The full workflow is:

```
raw data -> cleaning -> wrangling -> graph construction -> PageRank -> feature
engineering -> EDA -> train/test split -> modeling -> diagnostics -> lifted signed
PageRank -> results
```

Script	Purpose
02_clean_data.py	Parses raw text data, cleans vote records, removes unusable rows, and creates support edge list.
03_build_graph.py	Builds the directed support graph, computes PageRank, and saves graph summaries.
04_compute_features.py	Creates edge-level graph features and labels each edge as PageRank-uphill or non-uphill.
05_make_figures.py	Generates standard exploratory figures before modeling.
06_model_edges.py	Trains the baseline, logistic regression, and random forest models; saves metrics and confusion matrix.
07_advanced_figures.py	Creates ROC curve, precision-recall curve, feature importance, and uphill share by year.
08_signed_pagerank_extension.py	Computes lifted signed PageRank, neutral PageRank, signed ranking tables, correlations, and signed-network figures.
run_pipeline.py	Executes all project scripts in order.

4. Methods

4.1 Data Cleaning and Wrangling

The cleaning script parses the raw SNAP text records into a structured table with source, target, vote, result, year, date, and text fields. The workflow removes malformed records, missing or blank source/target fields, and self-loops. The support-only graph is built from vote == 1 records, while the full cleaned vote table is retained for signed PageRank analysis using support, oppose, and neutral vote layers.

Cleaned Data Item	Value
Clean vote records	196,511
Support-only edges	139,272
Support votes	143,822
Oppose votes	40,637
Neutral votes	12,052

4.2 Support Graph and PageRank

The baseline graph is a directed support-vote graph where a directed edge from user *i* to user *j* means that user *i* supported user *j* in an RfA vote. PageRank is computed with damping factor $\alpha = 0.85$. In this context, PageRank measures support-network centrality: users are more central when they receive support from other structurally important users.

Graph Metric	Value
Nodes	10,004
Edges	139,272
Density	0.001392
Self-loops	0
Weakly connected components	23
Strongly connected components	7,778

4.3 PageRank-Uphill Labeling and Feature Engineering

Each support edge is labeled as uphill if the target user has higher support-only PageRank than the source user. The engineered predictors include support_count, source and target in-degree, source and target out-degree, degree differences, and reciprocity. PageRank itself and PageRank difference are excluded from the predictive feature set to avoid direct target leakage.

Feature	Interpretation
support_count	Number of observed support votes for the aggregated source-target pair.
source_in_degree / source_out_degree	Support received and sent by the source voter.
target_in_degree / target_out_degree	Support received and sent by the target candidate.
in_degree_diff	target_in_degree - source_in_degree.
out_degree_diff	target_out_degree - source_out_degree.
reciprocal	Indicator for whether the target also has an edge back to the source.

4.4 Data Exploration

Exploratory analysis examines cleaned vote counts, graph size, PageRank distributions, degree distributions, PageRank edge differences, class balance, and the temporal share of PageRank-uphill support edges. These summaries verify that the graph is sparse, highly skewed, and suitable for both network analysis and edge-level classification.

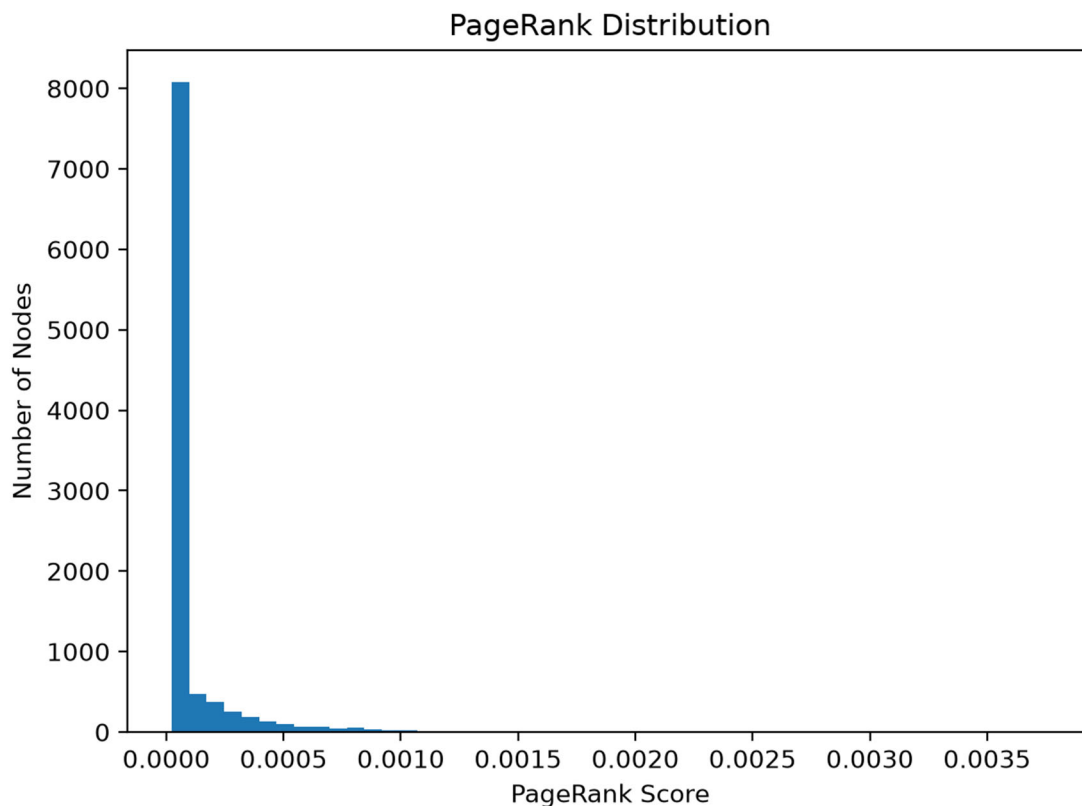


Figure 1. Distribution of support-only PageRank scores across users.

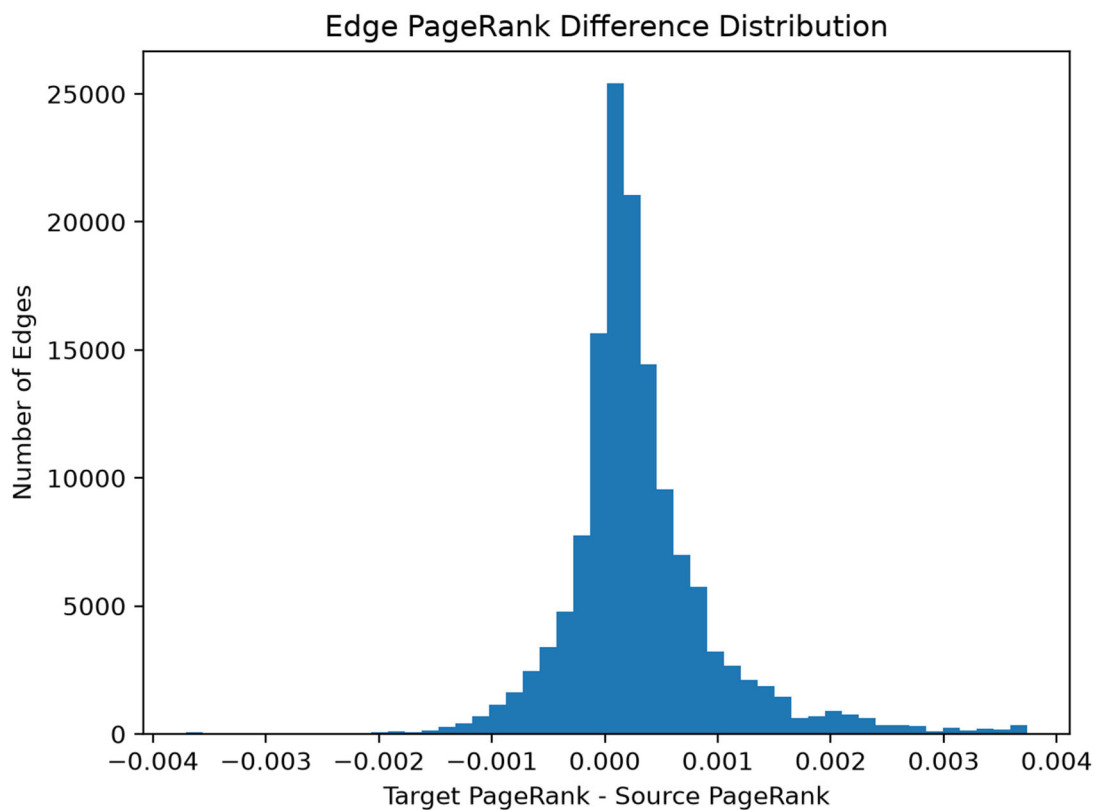


Figure 2. Distribution of $\text{PageRank}(\text{target})$ minus $\text{PageRank}(\text{source})$ across support edges.

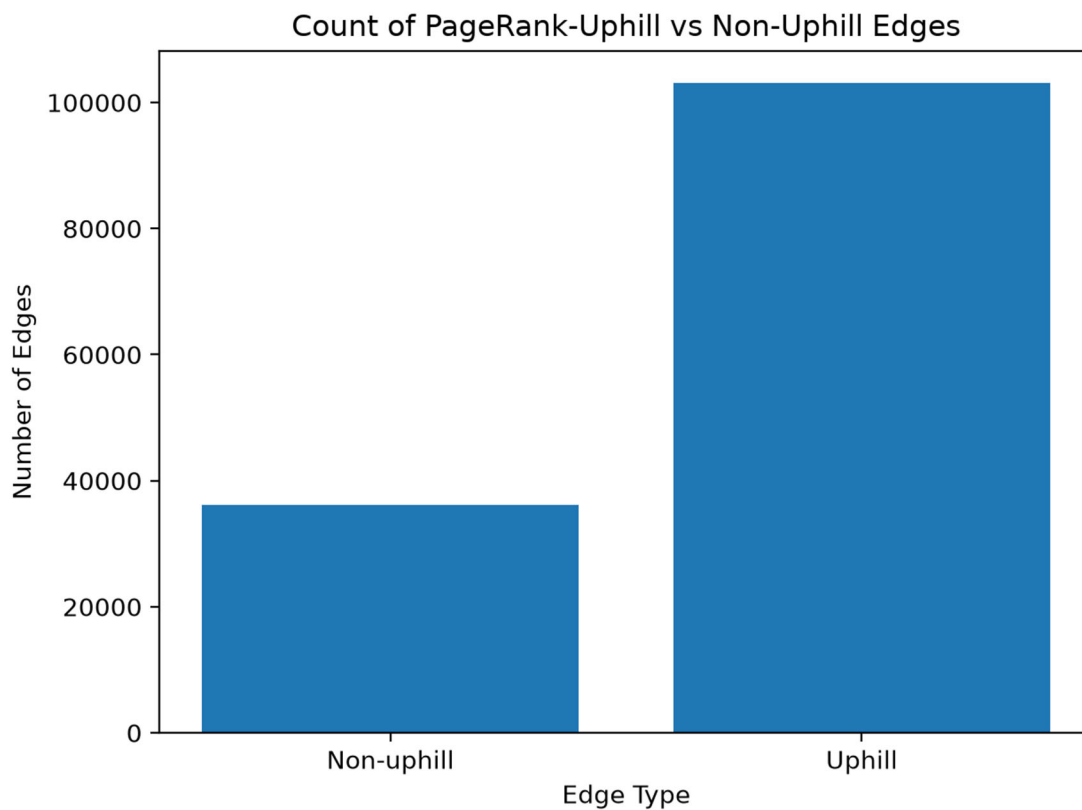


Figure 3. PageRank-uphill versus non-uphill support-edge counts.

4.5 Train/Test Split and Modeling

The supervised modeling stage uses stratified 80/20 train/test split with `random_state = 42` and stratification on the uphill target. Stratification is appropriate because PageRank-uphill edges are much more common than non-uphill edges. The project compares a majority-class baseline, logistic regression, and a random forest classifier.

Model	Role
Majority-class baseline	Reference model that always predicts the most common class.
Logistic regression	Interpretable linear classifier for graph-feature effects.
Random forest	Nonlinear classifier capable of capturing feature interactions.

4.6 Lifted Signed PageRank

The signed-network component incorporates support, oppose, and neutral votes without placing negative probabilities into ordinary PageRank. Each user is represented by a positive and negative reputation state. Support edges preserve sign, while oppose edges flip sign. This produces a nonnegative Markov chain on a doubled state space and allows PageRank-style computation of positive and negative reputation mass. Neutral votes are modeled separately as a neutral-attention PageRank layer.

Signed Score	Interpretation
<code>pagerank_support</code>	Ordinary PageRank on the support layer over the all-vote user universe.
<code>pagerank_oppose</code>	Ordinary PageRank on the oppose layer.
<code>pagerank_neutral</code>	Ordinary PageRank on the neutral-attention layer.
<code>signed_positive</code>	Positive reputation mass from lifted signed PageRank.
<code>signed_negative</code>	Negative reputation mass from lifted signed PageRank.
<code>signed_attention</code>	<code>signed_positive + signed_negative</code> .
<code>signed_net</code>	<code>signed_positive - signed_negative</code> .
<code>signed_balance</code>	<code>signed_net / signed_attention</code> .
<code>rank_shift_signed_minus_support</code>	Change from support-only rank to signed-net rank.

5. Results

5.1 Support-Only PageRank Results

Rank	Node	PageRank
1	Everyking	0.003764
2	Legoktm	0.003655
3	SarahStierch	0.003604
4	HJ Mitchell	0.003343
5	Mkdw	0.003285
6	TenPoundHammer	0.003160
7	Ironholds	0.002747
8	Tokyogirl79	0.002733
9	Drmies	0.002708
10	Werdna	0.002621

The top support-only PageRank users are structurally central recipients of support votes. The PageRank distribution is right-skewed, meaning that most users have small centrality values while a small group receives concentrated structural visibility.

5.2 PageRank-Uphill Edge Results

Edge Type	Count	Share
PageRank-uphill	103,065	74.00%
Non-uphill	36,207	26.00%

A large majority of support edges are PageRank-uphill: 103,065 of 139,272 support edges point from a lower-PageRank source to a higher-PageRank target. This provides the main empirical evidence for the project's domination-feasibility interpretation.

5.3 Machine-Learning Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Majority-class baseline	0.7400	0.7400	1.0000	0.8506	0.5000
Logistic regression	0.9207	0.9345	0.9601	0.9471	0.9705
Random forest	0.9251	0.9468	0.9524	0.9496	0.9783

The random forest is the strongest model by F1 score and ROC-AUC. The majority-class baseline has a high F1 score because the positive uphill class is common, but its ROC-AUC is 0.5000, confirming that it has no threshold-based discriminative power. Logistic regression and random forest both substantially outperform the baseline.

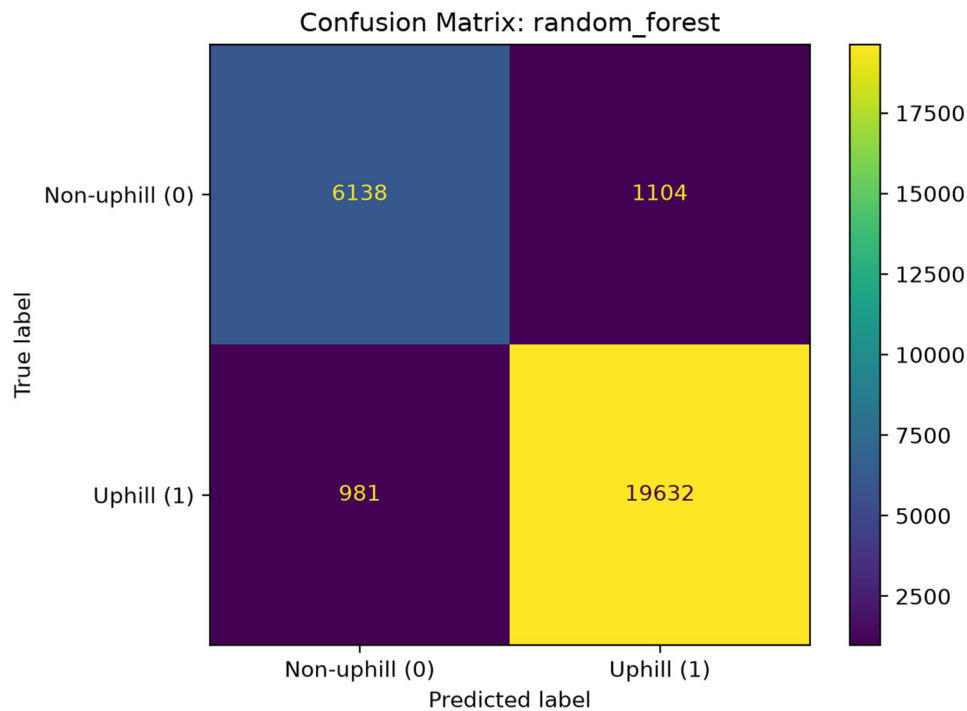


Figure 4. Random forest confusion matrix on the stratified test set.

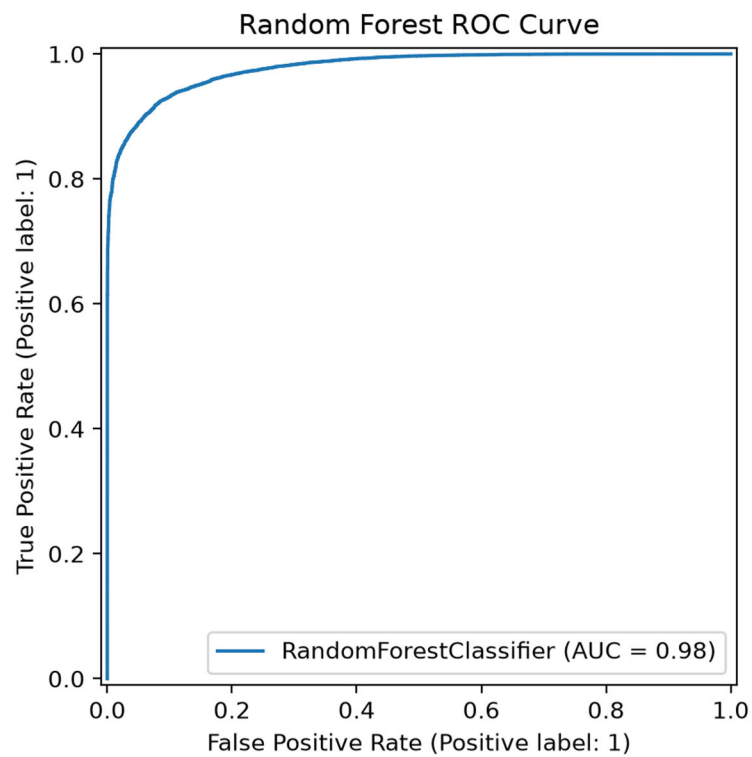


Figure 5. Random forest ROC curve for PageRank-uphill classification.

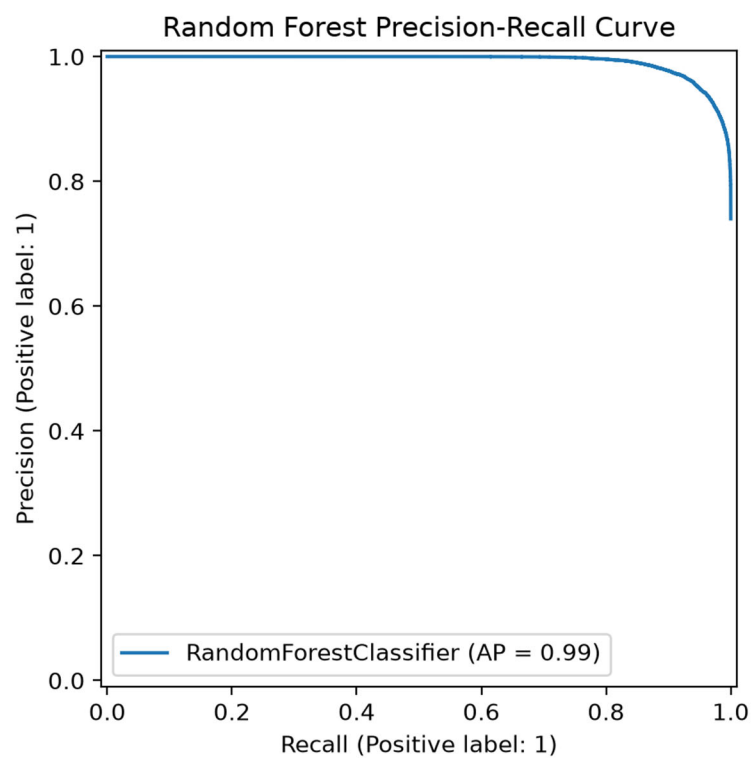


Figure 6. Random forest precision-recall curve under class imbalance.

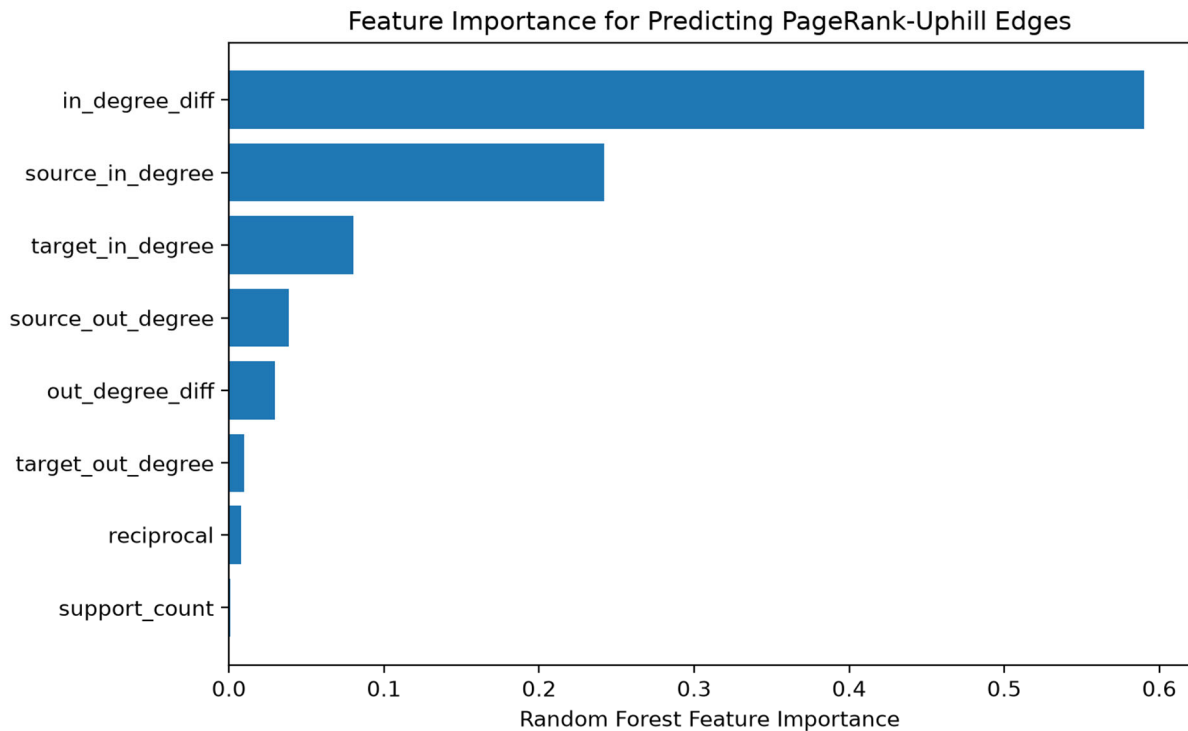


Figure 7. Random forest feature importance for engineered graph features.

5.4 Signed PageRank Results

Signed PageRank Item	Value
Users across all vote layers	11,377
Missing values in signed score table	0
Signed attention sum	1.000000000000
Minimum signed balance	-0.4573
Maximum signed balance	1.0000
Signed PageRank convergence	133 iterations
Pairwise signed-score correlations	90 rows

The signed PageRank extension passed numerical QA checks: the score table has no missing values, the signed-attention distribution sums to approximately 1.0, and the signed-balance values remain between -1 and 1. The 11,377 users represent the union of users appearing across support, oppose, and neutral vote layers.

Rank	Node	Signed Net	Signed +	Signed -	Support PR	Oppose PR
1	Legoktm	0.001030	0.002125	0.001095	0.003540	0.000052
2	SarahStierch	0.000978	0.001717	0.000738	0.003491	0.000049
3	HJ Mitchell	0.000926	0.001827	0.000901	0.003240	0.001297
4	Can't sleep, clown will eat me	0.000891	0.001379	0.000488	0.002094	0.000381
5	Drmies	0.000890	0.001448	0.000558	0.002624	0.000092
6	Mkdw	0.000837	0.001604	0.000768	0.003178	0.000576
7	Mark Arsten	0.000762	0.001300	0.000538	0.002290	0.000049
8	Dabomb87	0.000756	0.001143	0.000388	0.002079	0.000049
9	Werdna	0.000711	0.001462	0.000750	0.002546	0.001164
10	Elonka	0.000702	0.001630	0.000928	0.002181	0.002585

Node	Support Rank	Signed-Net Rank	Rank Shift	Signed Net	Support PR	Oppose PR
Everything	1	25	24	0.000545	0.003652	0.005078
Legoktm	2	1	-1	0.001030	0.003540	0.000052
SarahStierch	3	2	-1	0.000978	0.003491	0.000049
HJ Mitchell	4	3	-1	0.000926	0.003240	0.001297
Elonka	18	10	-8	0.000702	0.002181	0.002585

The signed-net ranking overlaps with the support-only ranking but changes the interpretation for users who also receive opposition. For example, Everything is ranked first by support-layer PageRank but lower by signed-net PageRank, while Legoktm becomes the top signed-net user. This shows that incorporating opposition and neutral attention reveals reputational structure that support-only PageRank cannot capture.

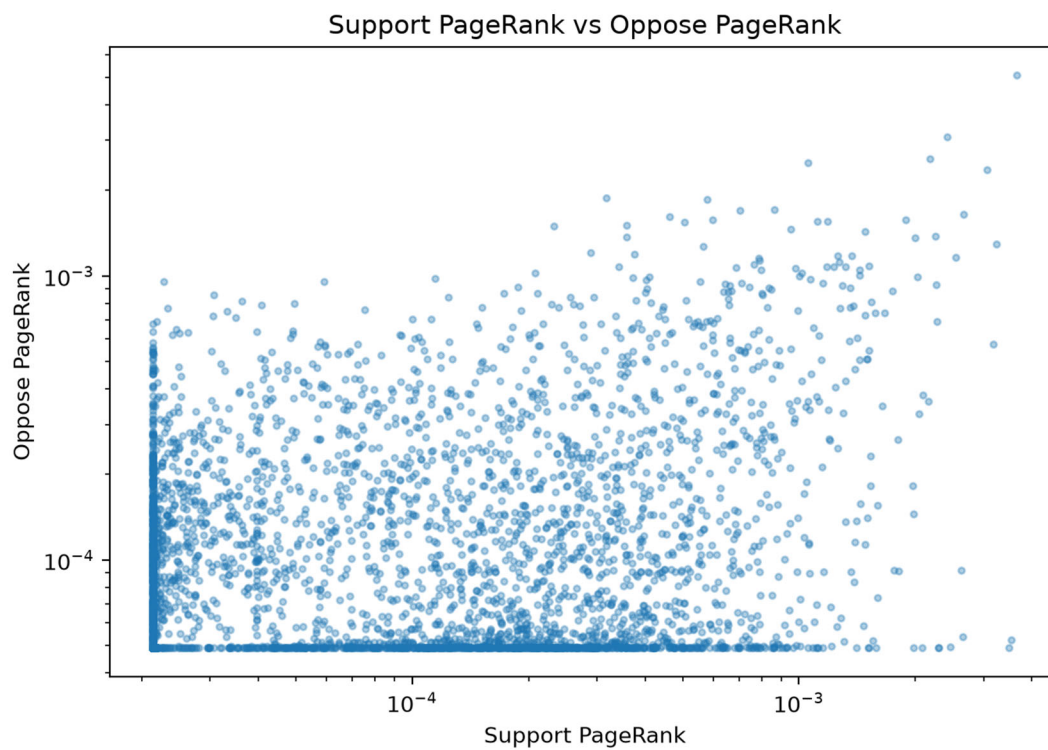


Figure 8. Support-layer PageRank versus oppose-layer PageRank.

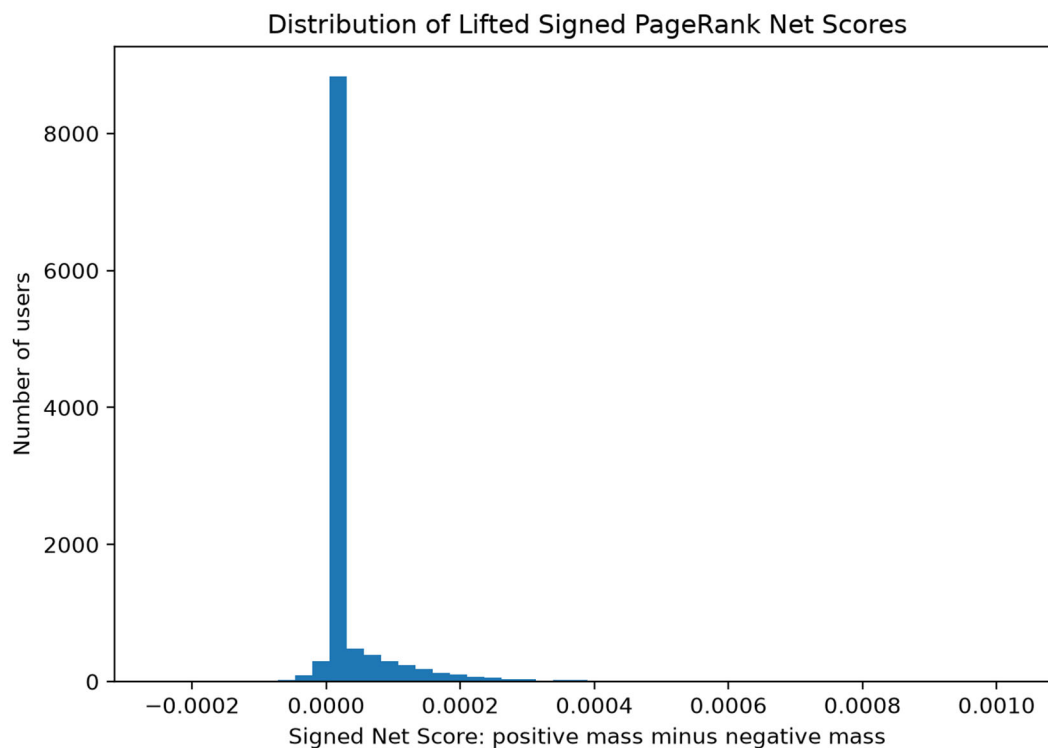


Figure 9. Distribution of signed-net reputation scores.

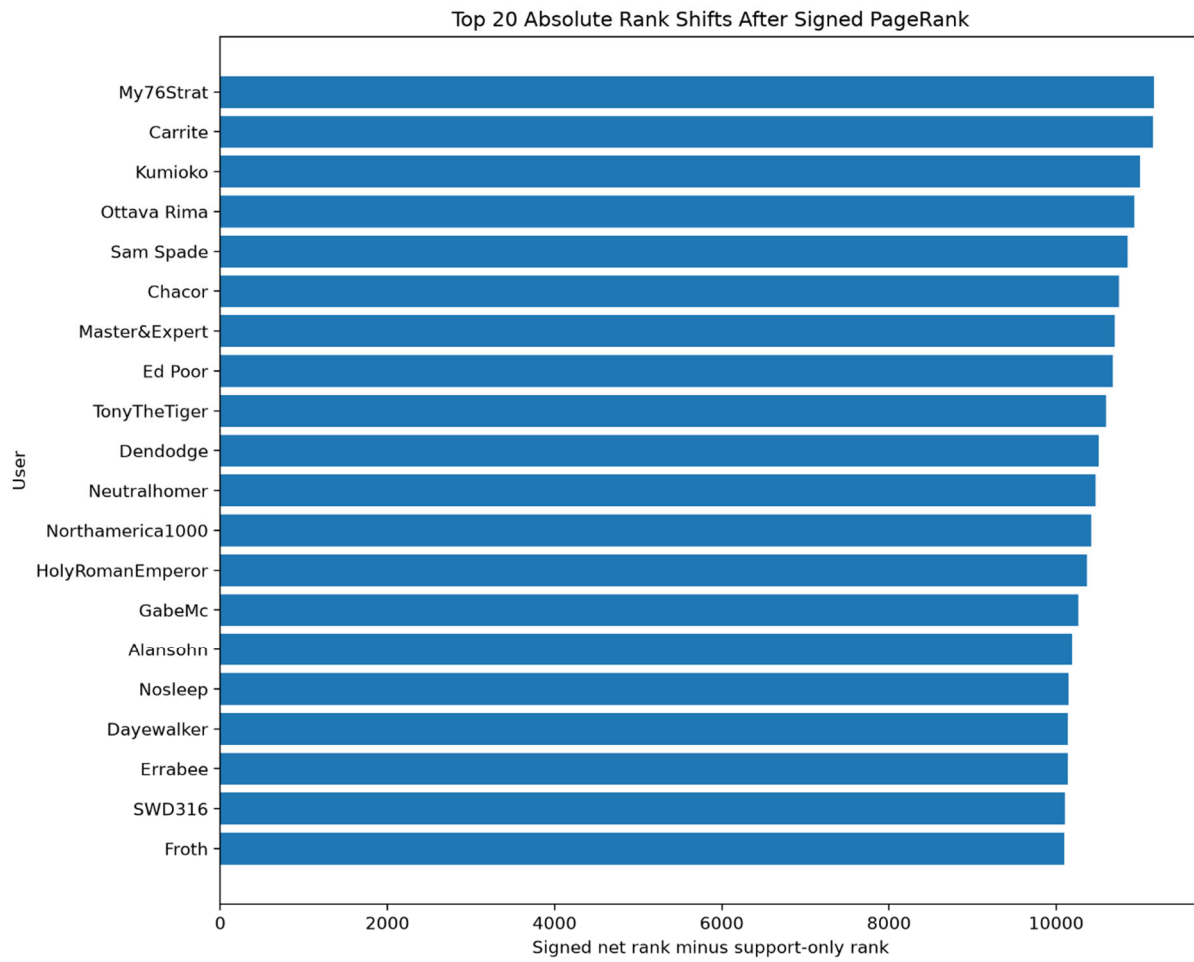


Figure 10. Largest rank shifts between support-only PageRank and signed-net PageRank.

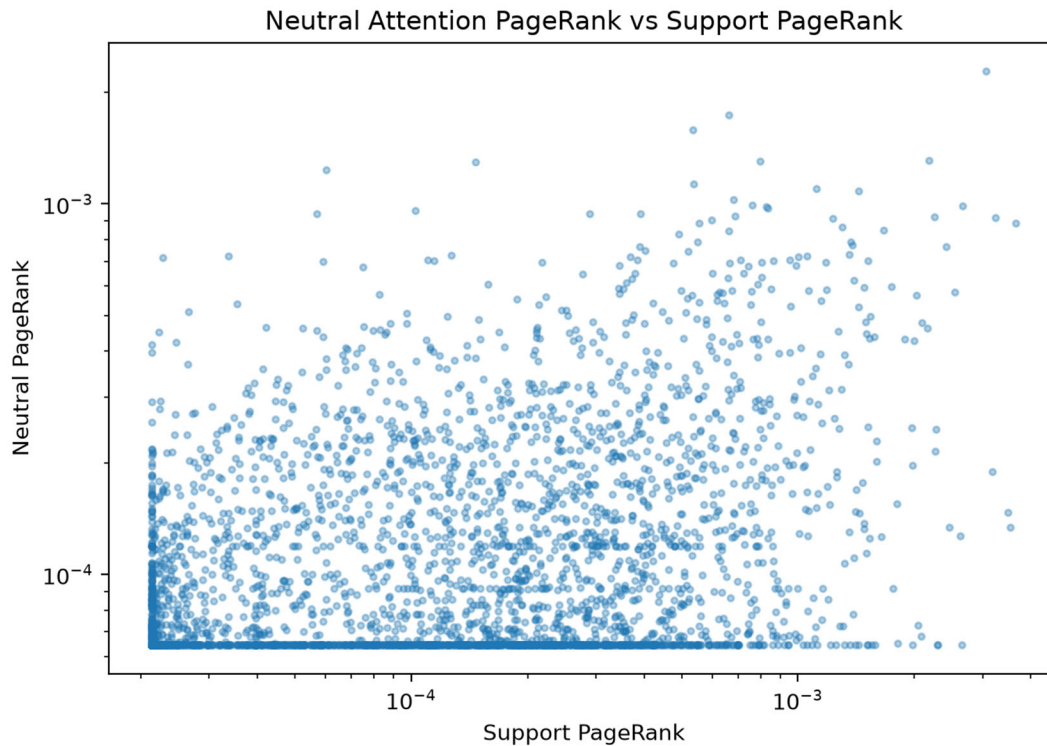


Figure 11. Neutral-attention PageRank versus support-layer PageRank.

5.5 Signed Score Correlations

Method	Score X	Score Y	Correlation
Pearson	pagerank_support	signed_positive	0.9821
Pearson	pagerank_support	signed_net	0.9212
Pearson	pagerank_support	pagerank_oppose	0.5005
Pearson	pagerank_oppose	signed_negative	0.7773
Pearson	pagerank_oppose	signed_net	0.2470
Pearson	pagerank_neutral	signed_net	0.3097
Spearman	pagerank_support	signed_positive	0.9451
Spearman	pagerank_support	signed_net	0.6065
Spearman	pagerank_oppose	signed_negative	0.8790

The correlation results show that support PageRank is strongly related to signed-positive mass and signed-net score, but not identical to them. Opposition PageRank is more closely related to signed-negative mass than to signed-net score. This supports the interpretation that the signed PageRank extension adds new information rather than simply duplicating the support-only PageRank ranking.

6. Results and Conclusions

The cleaned support-vote graph contains 10,004 nodes and 139,272 directed support edges. The raw vote table retained 196,511 usable vote records after cleaning, including 143,822 support votes, 40,637 oppose votes, and 12,052 neutral votes. The final support graph is sparse, with density 0.00139, and has 23 weakly connected components and 7,778 strongly connected components. This structure is consistent with a large social voting network in which many users participate locally or briefly while a smaller group occupies more central positions.

The central empirical finding is that PageRank-uphill support edges are common. Of the 139,272 support edges, 103,065 edges (74.00%) point from a lower-PageRank source to a higher-PageRank target. Only 36,207 edges (26.00%) are non-uphill. This means that support votes in the Wikipedia RfA network usually flow toward users who are more structurally central in the support graph. The exploratory figures reinforce this result: PageRank is right-skewed, target in-degree is

generally more concentrated than source in-degree, and the uphill-edge count plot shows a clear majority of PageRank-uphill support edges.

The machine-learning results show that uphill behavior is strongly predictable from local graph features. The majority-class baseline reaches F1 0.8506 because the uphill class is common, but its ROC-AUC is only 0.5000. Logistic regression improves to F1 0.9471 and ROC-AUC 0.9705. The random forest performs best with accuracy 0.9251, precision 0.9468, recall 0.9524, F1 0.9496, and ROC-AUC 0.9783. The ROC curve, precision-recall curve, confusion matrix, and feature-importance plot show that the random forest separates PageRank-uphill and non-uphill edges well while using graph-derived predictors rather than PageRank itself as an input.

The signed PageRank component provides a more holistic view of RfA centrality by incorporating support, oppose, and neutral votes. The lifted signed PageRank model converged in 133 iterations and produced a valid signed-attention distribution summing to approximately 1.0 with no missing score values. The signed-net ranking overlaps with the support-only ranking but changes the interpretation for users who receive substantial opposition. This demonstrates that centrality in the RfA network is multidimensional: once opposition and neutral votes are incorporated, the analysis distinguishes support centrality, opposition centrality, neutral attention, total signed attention, and net signed reputation.

Overall, the results support the project's PageRank feasibility interpretation. A strict one-sided domination condition requiring source nodes to dominate target nodes along support edges would not describe this real network well. Instead, the empirical pattern is the reverse for most edges: support frequently moves from less central voters toward more central candidates. The project therefore demonstrates how a theorem-inspired PageRank concept can be converted into an empirical diagnostic, then evaluated with reproducible graph analytics, feature engineering, visualization, supervised learning, and signed-network analysis.

7. Limitations and Next Steps

The project is best interpreted as an empirical audit of an observed historical network, not a causal model of Wikipedia governance. The SNAP dataset covers RfA votes from 2003 through May 2013, so the findings should not be assumed to describe current Wikipedia behavior. The results may also depend on graph-construction choices, including whether duplicate source-target pairs are aggregated, whether PageRank is weighted or unweighted, how dangling nodes are handled, and how neutral votes are interpreted.

The classifier is an edge-level model on an observed graph. A stricter future-prediction design would compute PageRank and graph features on a training graph only, then evaluate on temporally held-out edges. Future work could add PageRank damping-factor sensitivity checks, compare weighted and unweighted PageRank, use text-derived vote-comment features, model RfA outcomes directly, or use a temporal split where early RfA behavior predicts later edge patterns. Future signed-network work could compare the lifted signed PageRank approach used here against separate-layer PageRank, alternative signed random walks, and signed link-prediction methods.

8. Tech Stack and Reproducibility

Tool	Role in Project
Python	Main programming language.
pandas	Data parsing, cleaning, aggregation, feature table construction, and result summarization.
NumPy	Numerical support.
SciPy	Sparse matrix construction for lifted signed PageRank.
NetworkX	Directed graph construction and ordinary PageRank computation.
scikit-learn	Train/test split, baseline model, logistic regression, random forest, metrics, ROC curve, precision-recall curve, and confusion matrix.
Matplotlib	Data visualization.
Docker	Reproducible computing environment.
Git/GitHub	Version control and final delivery.

The local workflow is:

```
pip install -r requirements.txt
python src/run_pipeline.py
```

The Docker workflow is:

```
docker build -t pagerank-rfa-audit .
docker run --rm -it -v "${PWD}:/app" -w /app pagerank-rfa-audit
```

The Docker run executes all scripts from data cleaning through signed PageRank extension and produces processed data, outputs, and figures.

9. Expected Output Files

Output	Description
data/processed/rfa_votes_clean.csv	Cleaned vote-level data.
data/processed/rfa_support_edges.csv	Aggregated support-only edge list.
data/processed/edge_features.csv	Model-ready edge-level feature table.
outputs/graph_summary.json	Support graph summary statistics.
outputs/pagerank_scores.csv	Support-only PageRank scores.
outputs/top_25_pagerank_nodes.csv	Top support-only PageRank users.
outputs/model_metrics.csv	Model performance metrics.
outputs/signed_pagerank_scores.csv	Lifted signed PageRank and layer scores.
outputs/signed_pagerank_top_nodes.csv	Top users across signed centrality categories.
outputs/signed_score_correlations.csv	Pearson and Spearman correlations among signed scores.
figures/*.png	EDA, model-diagnostic, temporal, and signed-network figures.

10. LLM Assistance Statement

I used ChatGPT as a coding, debugging, and writing-support tool to help design the project structure. It was also useful for troubleshooting Docker/GitHub issues, reviewing code, interpreting outputs, and preparing the written documentation. All final analysis decisions, code execution, interpretation, and submission materials were reviewed and edited by me.

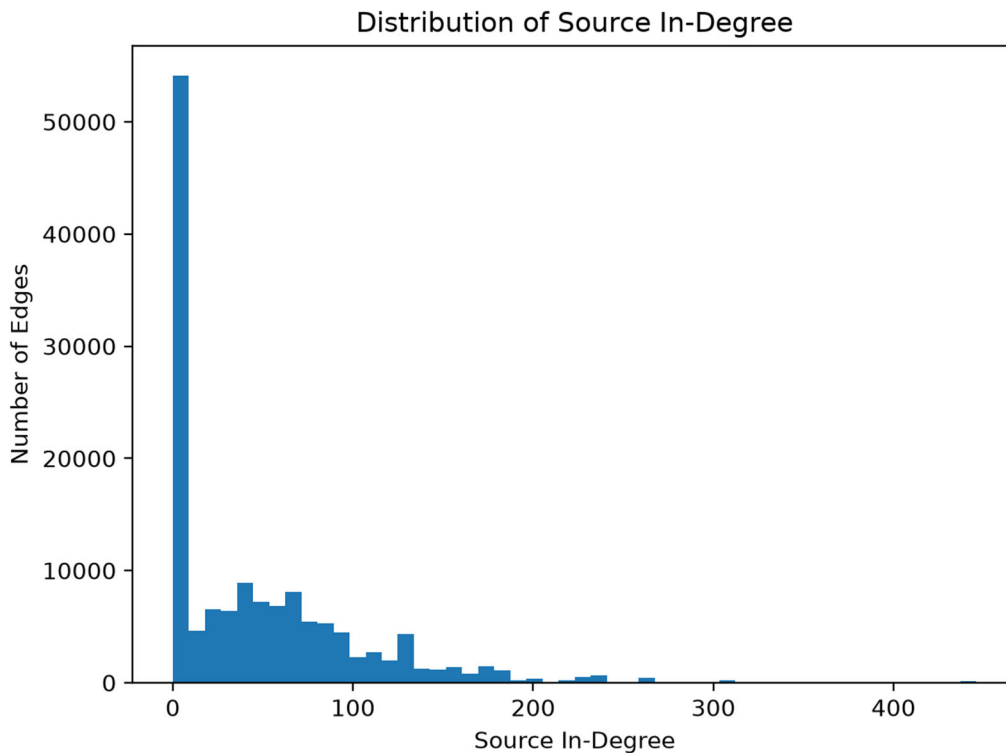
References

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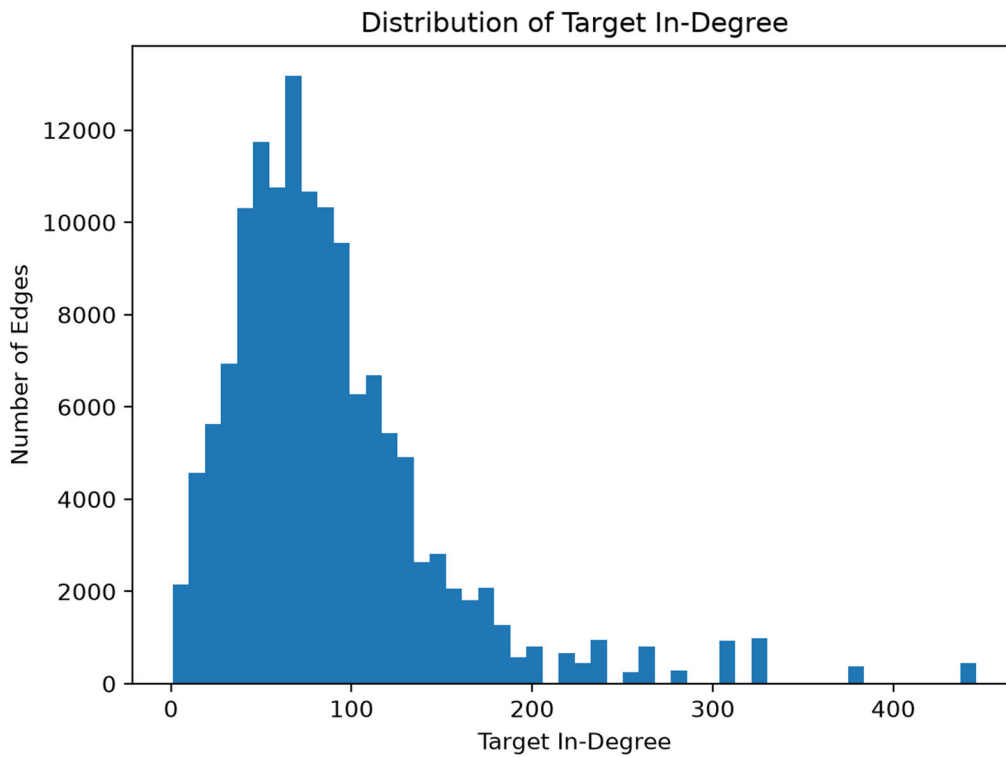
Appendix A. Complete Figure Catalog

The complete figure set produced by the pipeline is listed below. Selected figures are shown in the main report sections above.

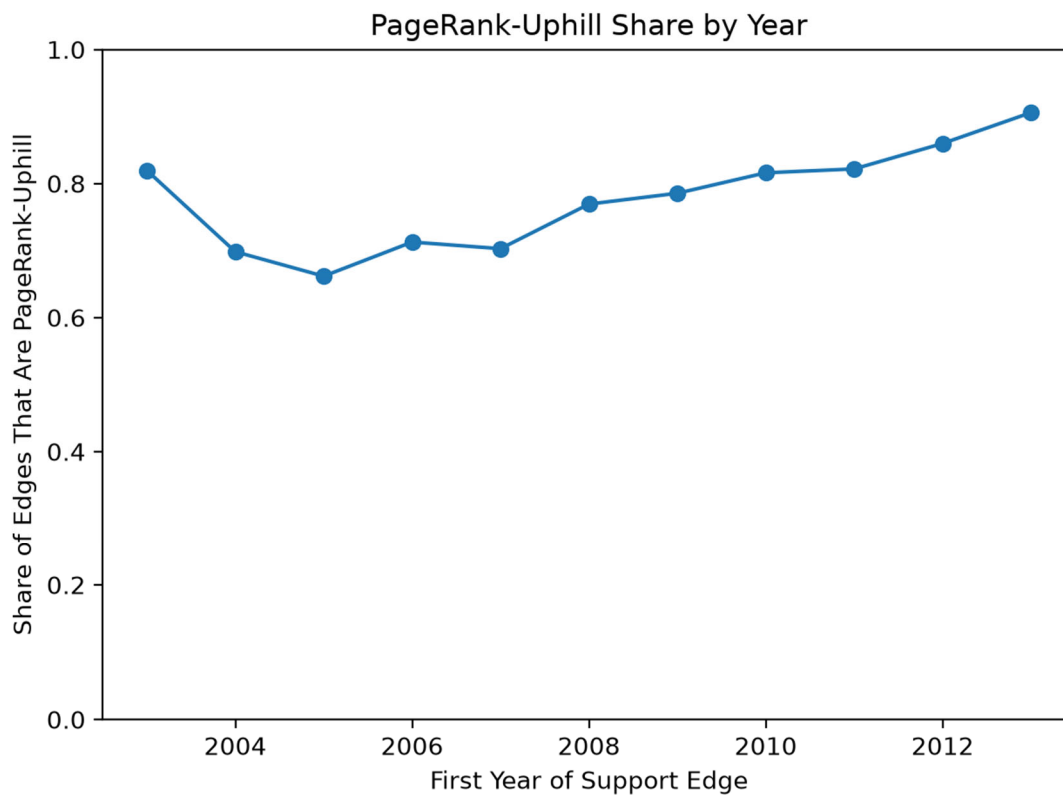
Figure File	Description
confusion_matrix.png	Random forest confusion matrix.
neutral_vs_support_pagerank.png	Neutral-attention PageRank versus support-layer PageRank.
pagerank_distribution.png	Distribution of support-only PageRank scores.
pagerank_edge_difference.png	Distribution of PageRank(target) minus PageRank(source).
precision_recall_curve_random_forest.png	Precision-recall curve for the random forest.
random_forest_feature_importance.png	Feature importance for random forest model.
roc_curve_random_forest.png	ROC curve for random forest model.
signed_net_score_distribution.png	Distribution of signed-net scores.
signed_rank_shift_top20.png	Largest rank shifts between support and signed-net PageRank.
signed_support_vs_oppose.png	Support-layer PageRank versus oppose-layer PageRank.
source_in_degree_distribution.png	Source in-degree distribution.
target_in_degree_distribution.png	Target in-degree distribution.
uphill_edge_counts.png	PageRank-uphill and non-uphill edge counts.
uphill_share_by_year.png	Uphill edge share by year.



Appendix Figure A1. Source in-degree distribution.



Appendix Figure A2. Target in-degree distribution.



Appendix Figure A3. PageRank-uphill edge share by year.